PROJECT

STREAMLINING TICKET ASSIGNMENT FOR EFFICIENT SUPPORT OPERATIONS

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PROJECT REPORT

INTRODUCTION

IDEATHON PHASE

REQUIREMENT ANALYSIS

PROJECT DESIGN PHASE

PROJECT PLANNING PHASE

PROJECT DEVELOPMENT PHASE

ADVANTAGES & DISADVANTAGES

PROJECT DOCUMENTATION

CONCLUSION

PROBLEM STATEMENT

ABC Corporation, a leading technology company, was facing challenges with efficiently assigning support tickets to the appropriate teams. With a vast array of products and services, the support team found it increasingly difficult to manually route tickets to the right groups, leading to delays in issue resolution and customer dissatisfaction.

OBJECTIVE

The objective of this initiative is to implement an automated system for ticket routing at ABC Corporation, aimed at improving operational efficiency by accurately assigning support tickets to the appropriate teams. This solution aims to reduce delays in issue resolution, enhance customer satisfaction, and optimize resource utilization within the support department.

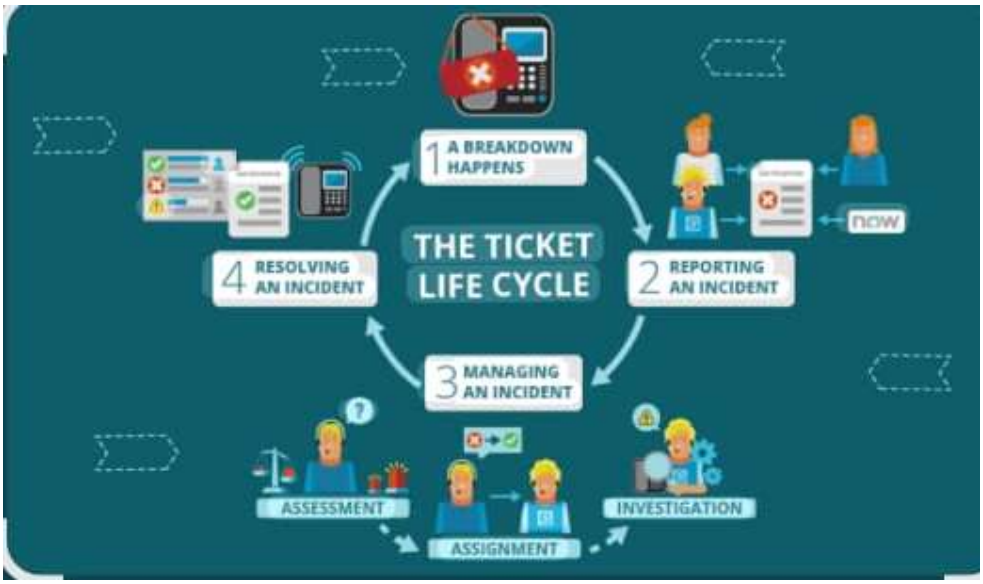
INTRODUCTION

In modern organizations, customer support plays an important role in maintaining service quality and customer satisfaction. Support teams receive numerous requests, complaints, and technical issues from users daily. These requests are usually managed through a ticketing system, where each issue is recorded as a support ticket.

However, in many organizations ticket assignment is still performed manually. Manual assignment often leads to delays, incorrect routing of issues, and inefficient handling of customer problems. This reduces productivity and increases response time.

The proposed system, Streamlining Ticket Assignment for Efficient Support Operations, aims to automate the process of ticket classification and assignment. The system ensures that each ticket is assigned to the appropriate support agent based on issue type, priority level, and agent availability.

By implementing an automated ticket assignment system, organizations can improve service efficiency, reduce response time, and enhance customer satisfaction.



IDENTIFICATION PHASE

The identification phase is the first stage in the development of the project. In this phase, the problem is recognized and the need for a solution is identified.

Organizations that handle large numbers of customer support requests often face challenges such as delayed responses, incorrect ticket assignments, and lack of proper tracking systems. These problems highlight the need for an automated system that can manage support tickets efficiently.

The goal of this project is to develop a system that streamlines ticket assignment and ensures that support requests are handled by the appropriate technician without unnecessary delays.

Objectives of the Project:

- Automate ticket assignment
- Improve response time
- Reduce manual workload
- Improve support service efficiency

Scope of the Project:

The system will allow users to submit support requests, automatically categorize them, assign them to support agents, and track the resolution process.

REQUIREMENT ANALYSIS

Requirement analysis is a crucial stage in which the needs of the users and the system are identified and documented. This process helps ensure that the final system meets user expectations.

Requirements are divided into two main categories: functional and non-functional requirements.

Functional Requirements

The system should perform the following functions:

- User registration and login
- Ticket submission by users
- Automatic ticket categorization
- Automatic assignment of tickets to support agents
- Ticket status tracking
- Admin dashboard
- Ticket closure and feedback

Non-Functional Requirements

Performance

The system should process ticket requests quickly without delays.

Security

User data and ticket information must be protected.

Reliability

The system should operate continuously without system failures.

Usability

The system interface should be easy to use for both customers and support agents.

PROJECT PLANNING PHASE

Project planning involves defining the timeline, resources, and tasks required to complete the project successfully. Proper planning helps ensure that the project is completed within the specified time.

During this phase, the project team prepares a project schedule and identifies the tools and technologies required for development.

Planning Activities:

- Define project scope
- Prepare development timeline
- Allocate project tasks
- Identify risks
- Select development tools

PROJECT PLANNING PHASE

Identify Problem

- Manual management of employee data is inefficient.

Define Objectives

- Develop an automated system.

Select Platform

- Chose the ServiceNow platform

Plan System Design

- Create forms, database tables, and workflows.

Test and Evaluate

- Ensure the system meets requirements.

Test and Evaluate

- Ensure the system meets requirements.

PROJECT DESIGN PHASE

The design phase converts the system requirements into a blueprint that guides developers during implementation.

System design includes the creation of system architecture, database structure, and interface design.

System Architecture

The system consists of the following components:

- User Interface
- Ticket Management Module
- Assignment Engine
- Database
- Notification System

Design Diagrams

1. Use Case Diagram

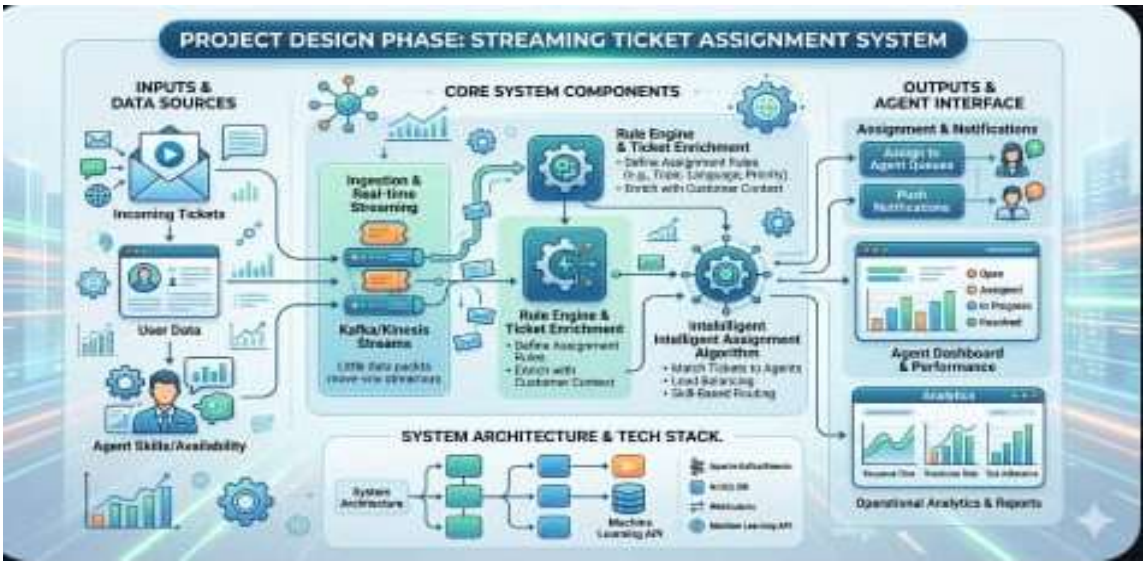
Shows interaction between users and the system.

2. Data Flow Diagram (DFD)

Represents how data flows within the system.

3. Entity Relationship Diagram (ERD)

Shows relationships between database tables.



PROJECT DEVELOPMENT PHASE

The development phase involves implementing the system using programming languages and development tools.

Developers write the code according to the design created in the previous stage. The system modules are developed and integrated to create the final application.

Development Modules:

- User Management Module
- Ticket Creation Module
- Ticket Assignment Module
- Notification Module
- Reporting Module

Technologies Used

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python / Java / PHP
Database	MySQL
Tools	VS Code, Git

During development, each module is tested to ensure proper functionality.

PROJECT DOCUMENTATION

Documentation is an essential part of the project development process. It records all details related to the design, development, and operation of the system.

Proper documentation helps developers understand the system and makes future maintenance easier.

Types of Documentation:

1. Project Proposal
2. Software Requirement Specification (SRS)
3. System Design Documents
4. User Manual
5. Technical Documentation
6. Final Project Report

Documentation also helps evaluators understand the project implementation during project assessment.

PROJECT DEMONSTRATION

Project demonstration is the final stage where the completed system is presented to the project guide or evaluation committee.

During the demonstration, the project team explains the objectives, system architecture, and working of the system.

Demonstration Steps:

1. Introduction of the project topic
2. Explanation of system design
3. Demonstration of ticket creation
4. Demonstration of automatic ticket assignment
5. Showing ticket tracking and resolution
6. Answering questions from the evaluation panel

The demonstration allows evaluators to verify that the system works as expected.

ADVANTAGES OF THE PROPOSED SYSTEM

The proposed system provides several benefits:

- Faster ticket assignment
- Reduced manual workload
- Improved customer satisfaction
- Better issue tracking
- Efficient support management

ADVANTAGES AND DISADVANTAGES

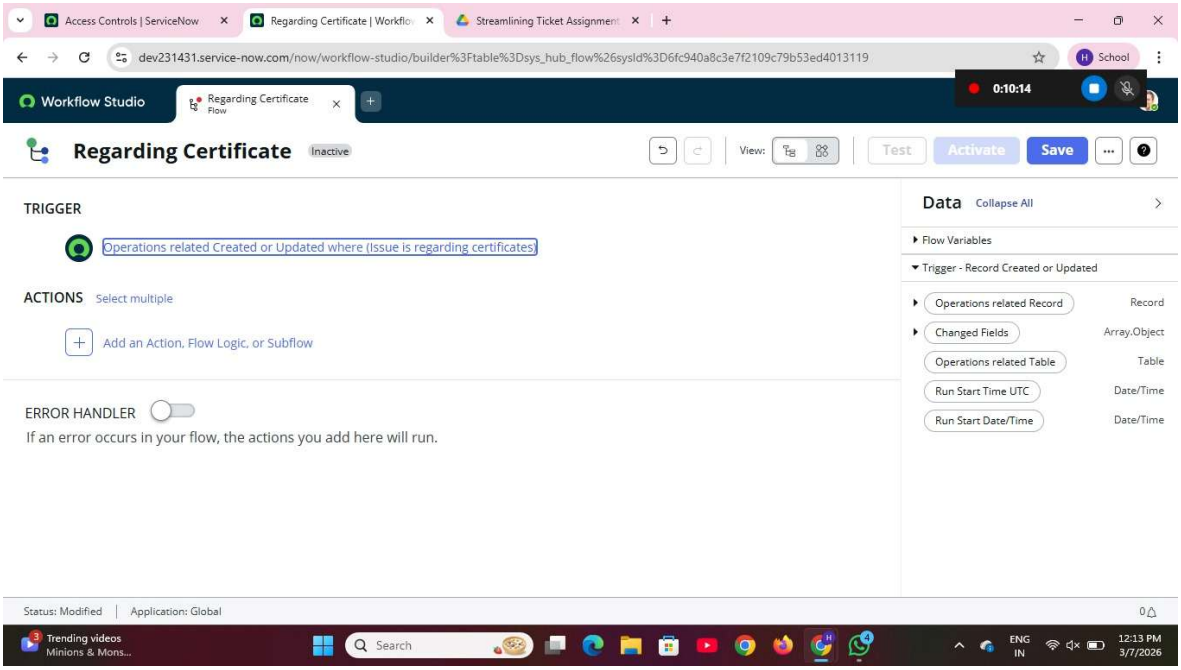
Advantages:

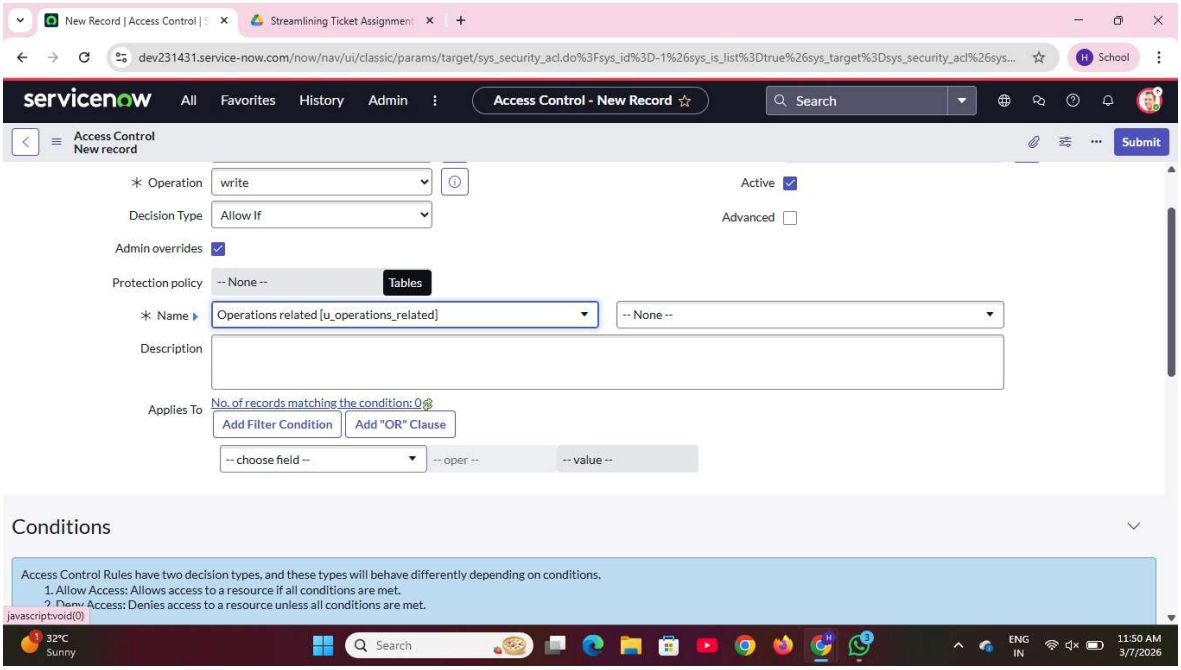
1. Faster Ticket Handling – Automatically assigns tickets to support agents, reducing response time.
2. Improved Efficiency – Reduces manual work in assigning tickets.
3. Better Resource Management – Distributes workload evenly among support agents.
4. Improved Customer Satisfaction – Quick response and faster issue resolution improve customer experience.
5. Easy Tracking – Helps monitor ticket status and progress easily.

Disadvantages:

1. Initial Setup Cost – Developing and implementing the system may require time and resources.
2. System Dependency – If the system fails, ticket assignment may be delayed.
3. Training Required – Support staff may need training to use the new system effectively.
4. Complexity – Automatic assignment rules may become complex in large organizations.

SCREENSHOTS




New Record | Access Control | S
Streamlining Ticket Assignment
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-- None --
Description
Applies To
No. of records matching the condition: 0
Add Filter Condition
Add "OR" Clause
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-- oper --
-- value --
Conditions
Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met.
2. Deny Access: Denies access to a resource unless all conditions are met.
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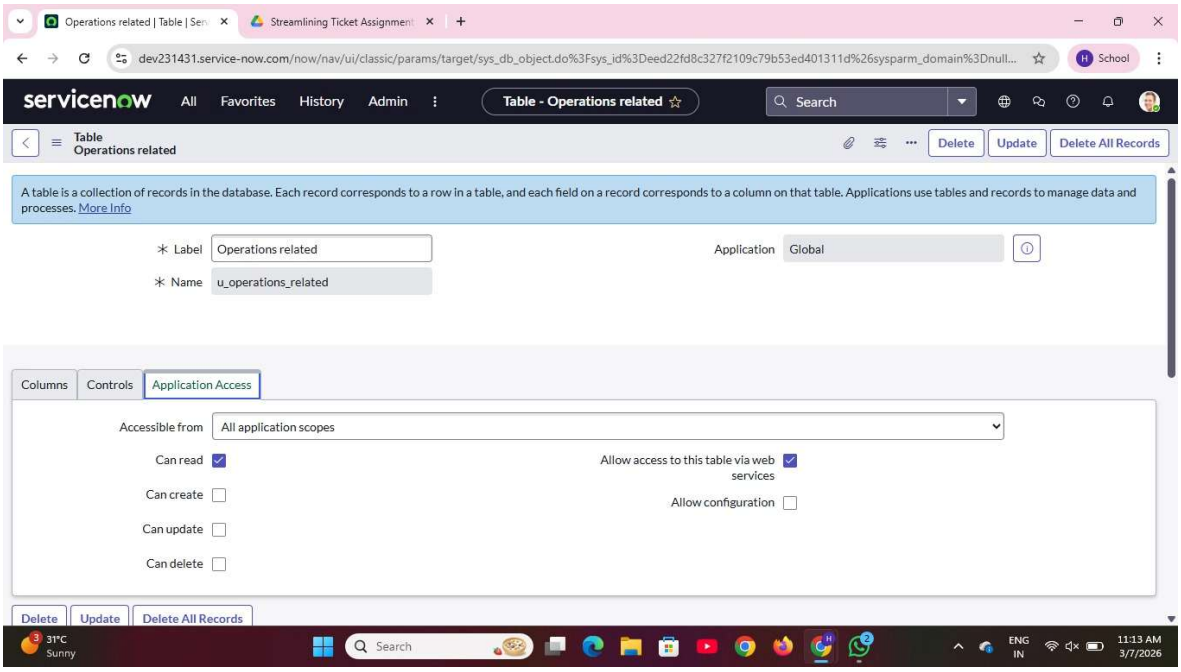
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Streamlining Ticket Assignment
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servicenow
All
Favorites
History
Admin
Table - Operations related
Search
Table
Operations related
Delete
Update
Delete All Records
A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. More Info
Label
Operations related
Application
Global
Name
u_operations_related
Columns
Controls
Application Access
Accessible from
All application scopes
Can read
Allow access to this table via web services
Can create
Allow configuration
Can update
Can delete
Delete
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FUTURE SCOPE

The project “Streamlining Ticket Assignment for Efficient Support Operations” can be further improved by adding advanced features and technologies in the future. These enhancements can make the system more intelligent, efficient, and user-friendly.

One possible improvement is the integration of Artificial Intelligence (AI) and Machine Learning (ML) to automatically classify tickets based on the issue description. This will help the system assign tickets more accurately to the appropriate support agents.

Another future enhancement is the implementation of predictive analytics, which can analyse previous ticket data to predict common issues and help organizations prepare solutions in advance.

The system can also be improved by adding real-time notifications through email or mobile applications so that support agents and customers receive updates about ticket status immediately.

Additionally, integrating the system with chatbots and automated support systems can help answer common customer queries without requiring human intervention, reducing the workload of support agents.

In the future, the system can also support cloud-based deployment, allowing organizations to manage support operations from anywhere and handle a larger number of tickets efficiently.

Overall, these improvements will make the ticket management system smarter, faster, and more effective in handling customer support operations.

Conclusion

The project Streamlining Ticket Assignment for Efficient Support Operations focuses on improving the efficiency of customer support systems. By automating the process of

ticket categorization and assignment, the system reduces manual intervention and ensures faster issue resolution.

The proposed solution enhances productivity, improves customer satisfaction, and provides better management of support operations. Implementing such systems in organizations can significantly improve service delivery and operational efficiency.

This project demonstrates how modern software technologies can be used to optimize support workflows and improve the overall service management process.

PROJECT PHASES OVERVIEW

IDEATION PHASE

- Identify the problem of manual management of employee data.
- Generate ideas for an automated solution.

REQUIREMENT ANALYSIS

- Define data flow diagrams and user stories.
- List functional and non-functional requirements.

PROJECT DESIGN PHASE

- Develop a proposed solution using the ServiceNow platform.
- Design solution architecture.

PROJECT PLANNING PHASE

- Organize tasks and set objectives.
- Determine steps needed for successful project completion.

PROJECT DEVELOPMENT PHASE

- Test the system to ensure it functions correrctly.
- Evaluate performance and reliability.

FINAL REPORT

- Complete a detailed report on the project.
- Summarize results and findings.
- Confirm the system's effectiveness.